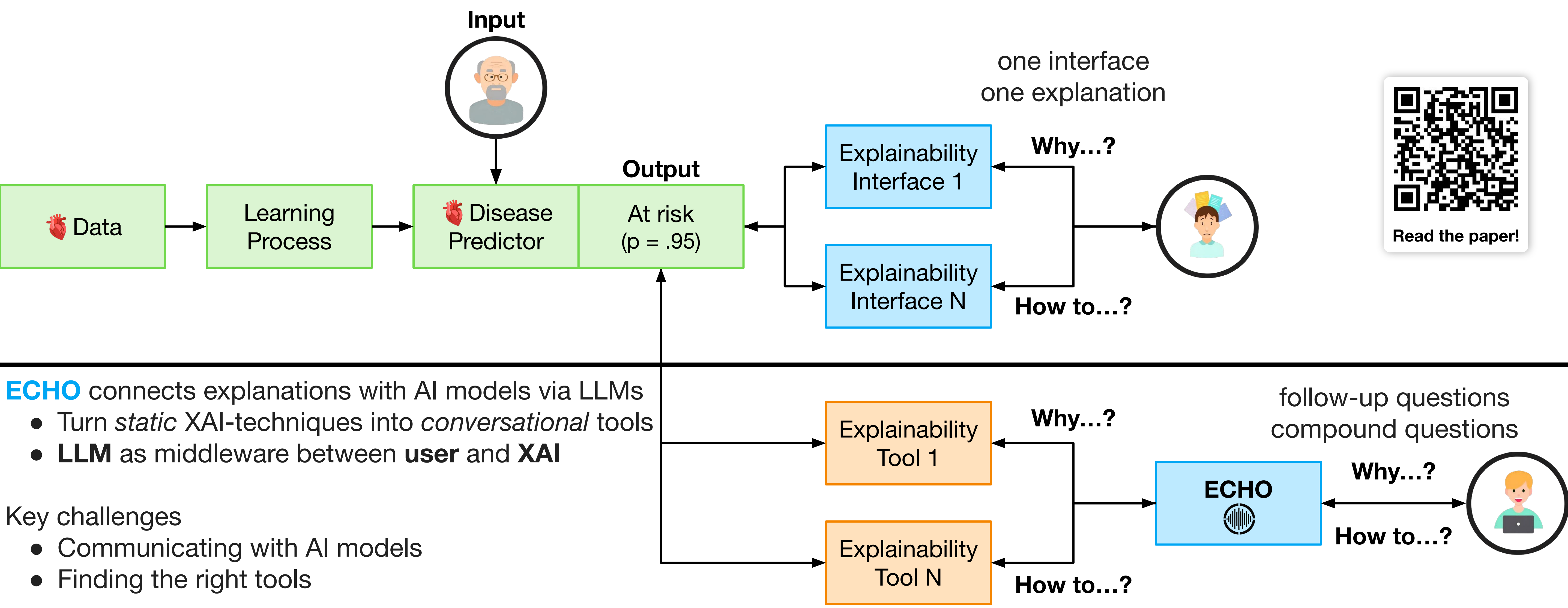


Enhancing Conversational Explainable AI through Tool-Augmented Language Models

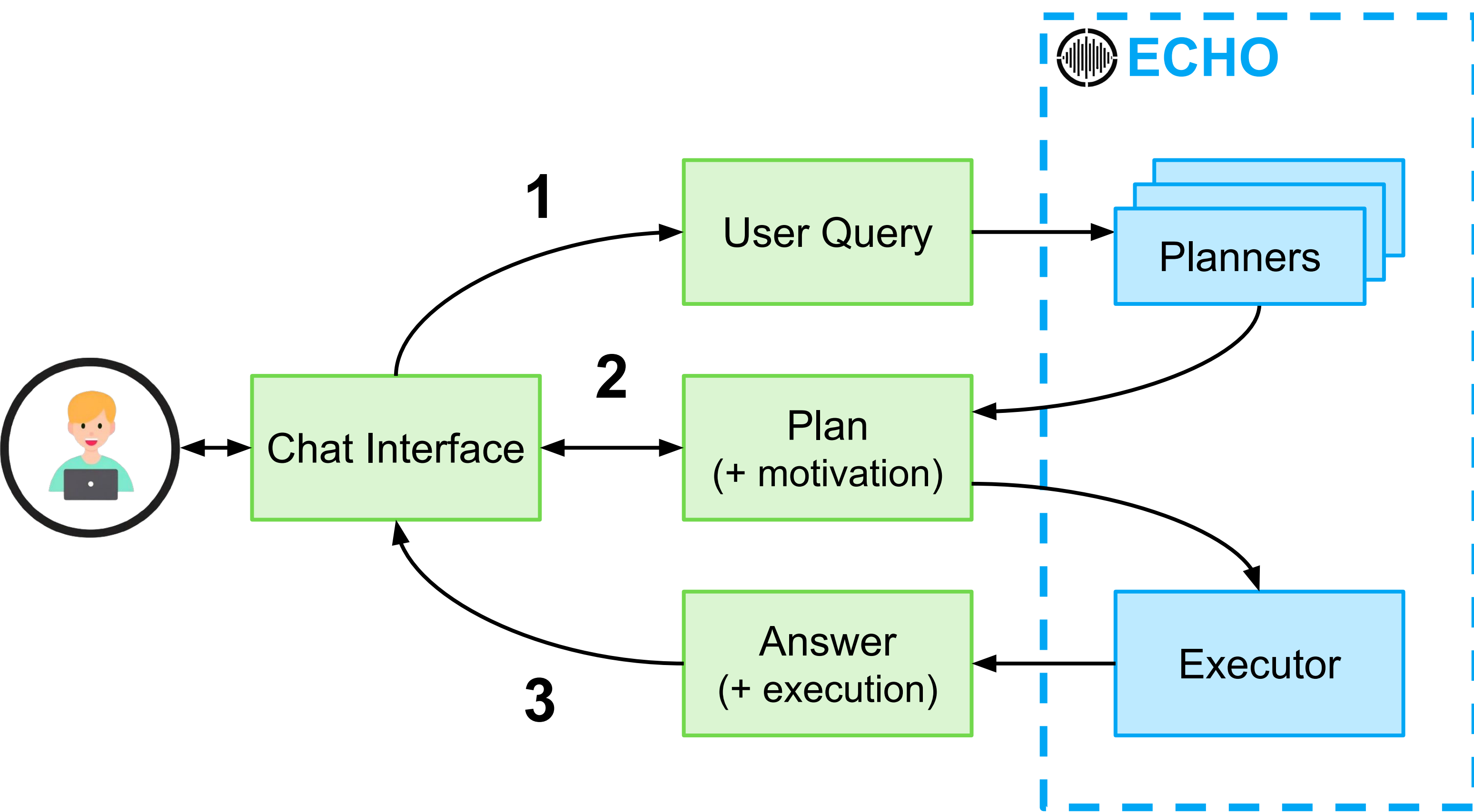
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From Static to Conversational eXplainable AI (XAI) with ECHO



ECHO's Internal Workings for Human-in-the-Loop Control



- ECHO** uses a controllable **multi-step mechanism**
1. User sends their **query** to a set of **planners**
 2. Each planner recommends a **plan**
 - A plan is a sequence of tools
 - Users can modify the final plan
 3. User executes their plan
 - The **executor** calls each tool in sequence
 - Final **answer** and execution trace sent to the user

What **tools** do we need for **communication with AI**?

Model-agnostic Tools

-  **Knowledge Retrieval:** generates additional relevant knowledge
-  **Inspector:** inspects implementation details in the source code
-  **Table Communicator:** queries the underlying tabular datasets
-  **Solution Generator:** uses CoT prompt to synthesize a final answer
-  **Answer Generator:** extracts the final answer from the solution

Model-specific Tools

- Challenge:** predefined tools are rigid
- They cannot handle model-specific functionality across tasks
 - Model specifications differ (e.g., prediction, feedback, interpretability)
- ECHO** uses **dynamic tool generation** to communicate with specific AI models
- Converts methods into tools (using *name*, *description*, *parameters*)
 - Enables richer control beyond prediction and explanation

ECHO in Case Studies

-  **Movie Recommender System**
 - Model: K-Nearest Neighbors
 - Type: White-box (interpretable)
 - Tools: Predict, Rate, LookupMovie
-  **Heart Disease Classifier**
 - Model: Decision Tree
 - Type: White-box (transparent)
 - Tools: Predict, GetTreeStructure
-  **Student Performance Classifier**
 - Model: Neural Network
 - Type: Black-box (opaque)
 - Tools: Predict, LIME, SHAP